PES University, Bengaluru

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE18CS352 - Cloud Computing

Total Marks : 100.0

1.a. List and explain the benefits of moving applications to the cloud. (4.0 Marks)

1.b. Explain the following terms with respect to Cloud Computing
a. Elasticity
b. Rapid provisioning
c. Measured Service
d. Resource Pooling (4.0 Marks)

1.c. Discuss the relative advantages and disadvantages of private and public clouds (6.0 Marks)

1.d. Explain different degrees of parallelism with an example (6.0 Marks)

2.a. Consider a situation where you are required to apply any one of these types of virtualization
Full Virtualization, Bare Metal virtualization, Host based virtualization and Para Virtualization
to different implementation technologies. Mark the appropriate virtualization type for each requirement and justify your answer.

(i) Run some dedicated applications on the VMs created on the guest OS and run
some other applications on the host OS directly
(ii) Run special APIs requiring substantial OS modifications in a VM
(iii) Run non-critical instructions on the hardware directly while critical instructions are discovered and replaced with traps into the VMM to be emulated by software.
(iv) Install the virtualization software directly on the hardware (8.0 Marks)

2.b. What is the difference between a hosted hypervisor and a bare metal hypervisor? Give an example of each (5.0 Marks)

2.c. What are controller-manager, kubelets and pods in Kubernetes? Explain with a diagram where each of them execute – on master or worker? (5.0 Marks)

2.d. What is the difference between hot migration and cold migration? (2.0 Marks)

3.a. Explain Lustre file system architecture with a neat diagram (8.0 Marks)

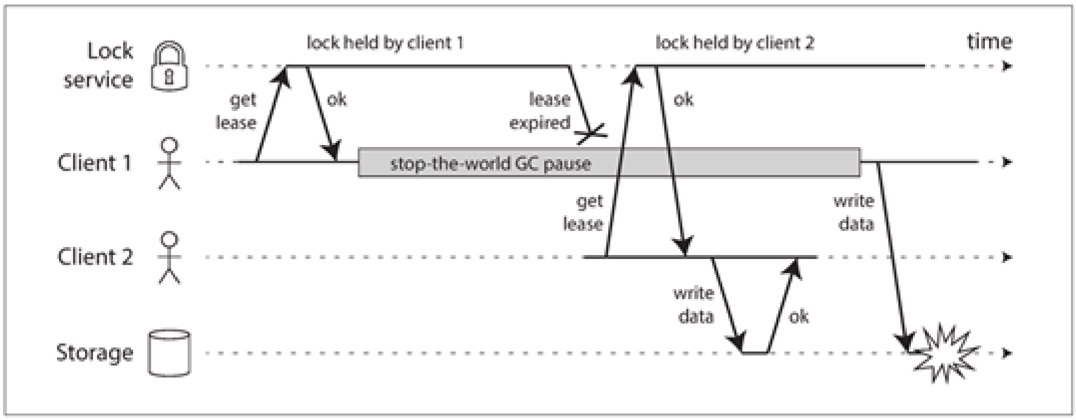
3.b. Briefly explain CAP theorem? (2 M) Explain how would you choose a database for your application based on CAP theorem (3M) (5.0 Marks)

3.c. What is replication in a distributed environment? Explain leaderless replication. (5.0 Marks)

3.d. What is rebalancing of partitions’? Why is it necessary to rebalance partitions? (2.0 Marks)

4.a. Explain the Ring Election algorithm with neat sketches. What are the problems associated with this algorithm? Discuss the worst-case scenario and messages required in worst case scenario. (8.0 Marks)

4.b. What is the problem with the implementation of a distributed lock in the following diagram? Explain with a diagram the approach that is used to overcome the problem.



(6.0 Marks)

4.c. How does Zookeeper work? What are the common services offered by Zookeeper ? (6.0 Marks)

5.a. Explain the following terms from Cloud Threat and Security Context. (2 Marks each)

1. Domain in Keystone
2. Token In Keystone
3. DoS Attack
4. Honeypot design pattern

(8.0 Marks)

5.b. What is Cloud Bursting? Explain how Cloud Bursting can be Beneficial to Cloud Users. (6.0 Marks)

5.c. How do you design a multitenant database with name-value pairs? Explain neatly with an example showing appropriate table entries

(6.0 Marks)